

I define Euclidean domain and prove every Euclidean domain is PID.

Definition 1. Let R be a domain. A norm on R is a function $N : R \rightarrow \mathbb{N}$ such that $N(0) = 0$.

Example 1. Let $R = \mathbb{Z}$. Then

1. $N(a) = |a|$ is a norm
2. $N'(a) = 0$ is a norm
3. $N''(a) = 1$ if $a \neq 0$ and $N''(0) = 0$ is a norm.

Definition 2. Let R be a domain. We say R is an Euclidean domain if there exists a norm $N : R \rightarrow \mathbb{N}$ such that for all $a, b \in R$ and $b \neq 0$, we get

$$a = bq + r$$

where $r = 0$ or $N(r) < N(b)$.

Example 2. In the setting of Example 1, $(\mathbb{Z}, N)$ is an Euclidean domain, but $(\mathbb{Z}, N')$ and $(\mathbb{Z}, N'')$ are not Euclidean domain. Because $3 = 2 \times 1 + 1$, we note first, $r = 1 \neq 0$ and second, $N'(2) = N'(1)$ and similarly $N''(2) = N''(1)$.

Proposition 1. Let R be an Euclidean domain. Then every ideal I in R is principle.

Proof. Let $I \subseteq R$ be an ideal. If $I = (0)$, then I is principle. If $I \neq (0)$, consider $I - \{0\} \neq \emptyset$. Note the set $\{N(x) : x \in I - \{0\}\} \subseteq \mathbb{N}$ has a least element. Then there exists some $0 \neq d \in I$ such that $N(d)$ has the minimal norm on the set $I - \{0\} \neq \emptyset$. In particular, $(d) \subseteq I$.

We claim $d \mid a$ for all $a \in I$, and hence $I = (d)$ is principle. Since R is an Euclidean domain, write $a = qd + r$ where $r = 0$ or $N(r) < N(d)$. Suppose $r \neq 0$. Then $N(r) < N(d)$. It follows $r = a - qd \in I - \{0\}$ and $N(r) < N(d)$ contradicting the minimality of norm of d on the set $I - \{0\} \neq \emptyset$. So $r = 0$ and hence $a = qd \in (d)$. Then $I \subseteq (d)$ and our claim is proved. $\square$